

Descriptives & Zero-order associations – HPT (Czech data)

Means/SDs, distributions, correlations, and school/class variation

HPT and Extremism project

2026-10-02

Contents

1 Purpose	1
1.1 1) Data & packages	1
1.2 2) Scoring: constructs & subscores	2
1.3 3) Sample overview	3
1.4 4) Descriptives (means, SDs, n, range)	4
1.5 5) Distributions (histograms, same axes where possible)	5
1.6 6) Zero-order correlations (student level)	7
1.7 7) Between-group variation: school/class ICCs	8
1.8 8) Quick sanity plots by class/school (optional)	10
1.9 9) Reproducibility appendix	12
1.9.1 Notes & interpretation pointers	13

1 Purpose

This file establishes transparent baseline patterns—distributions, scale descriptives, zero-order correlations, and between-group variation—before any modeling. It follows the HPT framework (POP/ROA/CONT) and scoring conventions used in prior work, but with an explicit **presentism reversal** so that **higher HPT scores reflect better contextualization/agent-sensitive reasoning**. FR-LF and KSA reflect ideological/authoritarian agreement; SDR reflects social-desirability responding. These descriptives document the empirical landscape your hypotheses build on. *Interpretation cheatsheet appears under each table/figure.*

1.1 1) Data & packages

```
# Core
library(dplyr); library(stringr); library(tidyr); library(purrr); library(readxl)
# Tables & plots
library(knitr); library(kableExtra); library(ggplot2); library(scales)
```

```

# Correlations
library(psych)          # corr.test
# Multilevel & ICC
library(lme4)           # lmer
source("scoring_helpers.R")

theme_set(theme_minimal(base_size = 11))

# Load the dataset created in 00_data-preparation
normalised_responses <- readRDS("student_responses.RDS")
stopifnot(exists("normalised_responses"))

all_data <- normalised_responses

# --- Presentism reversal & canonical composites ---
POP_rev_items <- paste0("POP", 1:3)
stopifnot(all(POP_rev_items %in% names(all_data)))

all_data <- all_data %>%
  mutate(
    across(all_of(POP_rev_items), ~ 5 - suppressWarnings(as.numeric(.)),
      .names = "{.col}_rev")
  )

# Ensure factors as in codebook and build unique class_id
all_data <- all_data %>%
  mutate(
    school_id   = as.factor(school_id),
    class_label = as.factor(class_label),
    school_level = as.factor(school_level),
    school_type = as.factor(school_type),
    region      = as.factor(region),
    gender      = as.factor(gender),
    class_id    = interaction(school_id, class_label, drop = TRUE)
  )

```

1.2 2) Scoring: constructs & subscores

```

# HPT (1-4): subscores and totals (explicitly using REVERSED POP)
hpt_pop_items_rev <- paste0("POP", 1:3, "_rev") # reversed presentism
hpt_roa_items     <- c("ROA1", "ROA2", "ROA3")
hpt_cont_items    <- c("CONT1", "CONT2", "CONT3")

```

```

# Knowledge (0-6 correct)
kn_items <- paste0("KN", 1:6)

# FR-LF mini (1-5 Likert): RD1-3 + NS1-3
frlf_rd <- paste0("RD", 1:3)
frlf_ns <- paste0("NS", 1:3)

# KSA-3 authoritarianism (1-5 Likert): 9 items
ksa_items <- c(paste0("A",1:3), paste0("U",1:3), paste0("K",1:3))

# SDR-5 (1-5 Likert): SDR2-SDR4 already reversed in the dataset per codebook
sdr_items <- paste0("SDR", 1:5)

# Build scores (HPT_TOTAL := CTX6 as primary; also keep HPT_TOT9 for reference)
dat <- all_data %>%
  mutate(
    HPT_POP_raw = scale_mean(., POP_rev_items, min_answered = 2),
    HPT_POP     = scale_mean(., hpt_pop_items_rev, min_answered = 2),
    HPT_ROA     = scale_mean(., hpt_roa_items, min_answered = 2),
    HPT_CONT    = scale_mean(., hpt_cont_items, min_answered = 2),
    HPT_CTX6    = rowMeans(cbind(HPT_POP, HPT_CONT), na.rm = FALSE),
    HPT_TOT9    = rowMeans(cbind(HPT_POP, HPT_CONT, HPT_ROA),
                          na.rm = FALSE),
    HPT_TOTAL   = HPT_TOT9,
    KN_TOTAL    = rowSums(dplyr::select(., all_of(kn_items)), na.rm = TRUE),
    FRLF_RD     = scale_mean(., frlf_rd, min_answered = 2),
    FRLF_NS     = scale_mean(., frlf_ns, min_answered = 2),
    FRLF_MINI   = scale_mean(., c(frlf_rd, frlf_ns), min_answered = 4),
    KSA_TOTAL   = scale_mean(., ksa_items, min_answered = 7),
    SDR_TOTAL   = scale_mean(., sdr_items, min_answered = 4)
  )

```

How to read:

- **HPT (HPT_TOTAL)** uses reversed POP by construction; higher = better contextualization.
- **HPT_CTX6** (POP_rev + CONT) is our **primary descriptive score**; **HPT_TOT9** (adds ROA) is reported for reference.
- **Knowledge (KN_TOTAL)**: 0-6 correct.

FR-LF/KSA/SDR follow codebook.

1.3 3) Sample overview

```

vars_context <- c("school_id", "class_label", "school_level", "school_type", "region", "gender", "history_grade")
n_raw <- nrow(all_data); n_anal <- nrow(dat)

```

```
kable(data.frame(
  N_raw = n_raw,
  N_after_scoring = n_anal,
  classes = dplyr::n_distinct(all_data$class_id),
  schools = dplyr::n_distinct(all_data$school_id)
), caption = "Sample counters (after loading and scoring).") %>% kable_styling(full_width = FALSE)
```

Table 1: Sample counters (after loading and scoring).

N_raw	N_after_scoring	classes	schools
293	293	20	10

1.4 4) Descriptives (means, SDs, n, range)

```
# Report both totals explicitly
desc_vars <- c("HPT_POP", "HPT_ROA", "HPT_CONT", "HPT_CTX6", "HPT_TOT9",
              "KN_TOTAL", "FRLF_RD", "FRLF_NS", "FRLF_MINI",
              "KSA_TOTAL", "SDR_TOTAL")

desc_tbl <- dat %>%
  summarise(across(all_of(desc_vars),
    list(n = ~sum(!is.na(.)),
         mean = ~mean(., na.rm=TRUE),
         sd = ~sd(., na.rm=TRUE),
         min = ~min(., na.rm=TRUE),
         max = ~max(., na.rm=TRUE)))) %>%
  pivot_longer(everything(),
    names_to = c("variable", ".value"),
    names_pattern = "([~_]+_[~_]+|^[~_]+)_(n|mean|sd|min|max)")

# Keep original order
desc_tbl$variable <- factor(desc_tbl$variable, levels = desc_vars)

kable(desc_tbl[order(desc_tbl$variable),], booktabs = TRUE, digits = 2,
      caption = "Scale descriptives (student-level).") %>%
  kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)
```

Table 2: Scale descriptives (student-level).

variable	n	mean	sd	min	max
----------	---	------	----	-----	-----

HPT_POP	287	2.97	0.64	1.33	4.00
HPT_ROA	287	2.80	0.68	1.00	4.00
HPT_CONT	287	2.71	0.74	1.00	4.00
HPT_CTX6	287	2.84	0.55	1.33	4.00
HPT_TOT9	287	2.83	0.49	1.67	3.89
KN_TOTAL	293	3.04	1.62	0.00	6.00
FRLF_RD	285	2.54	0.88	1.00	5.00
FRLF_NS	286	2.43	0.89	1.00	5.00
FRLF_MINI	284	2.48	0.75	1.00	5.00
KSA_TOTAL	283	2.86	0.62	1.00	5.00
SDR_TOTAL	283	3.02	0.63	1.00	4.60

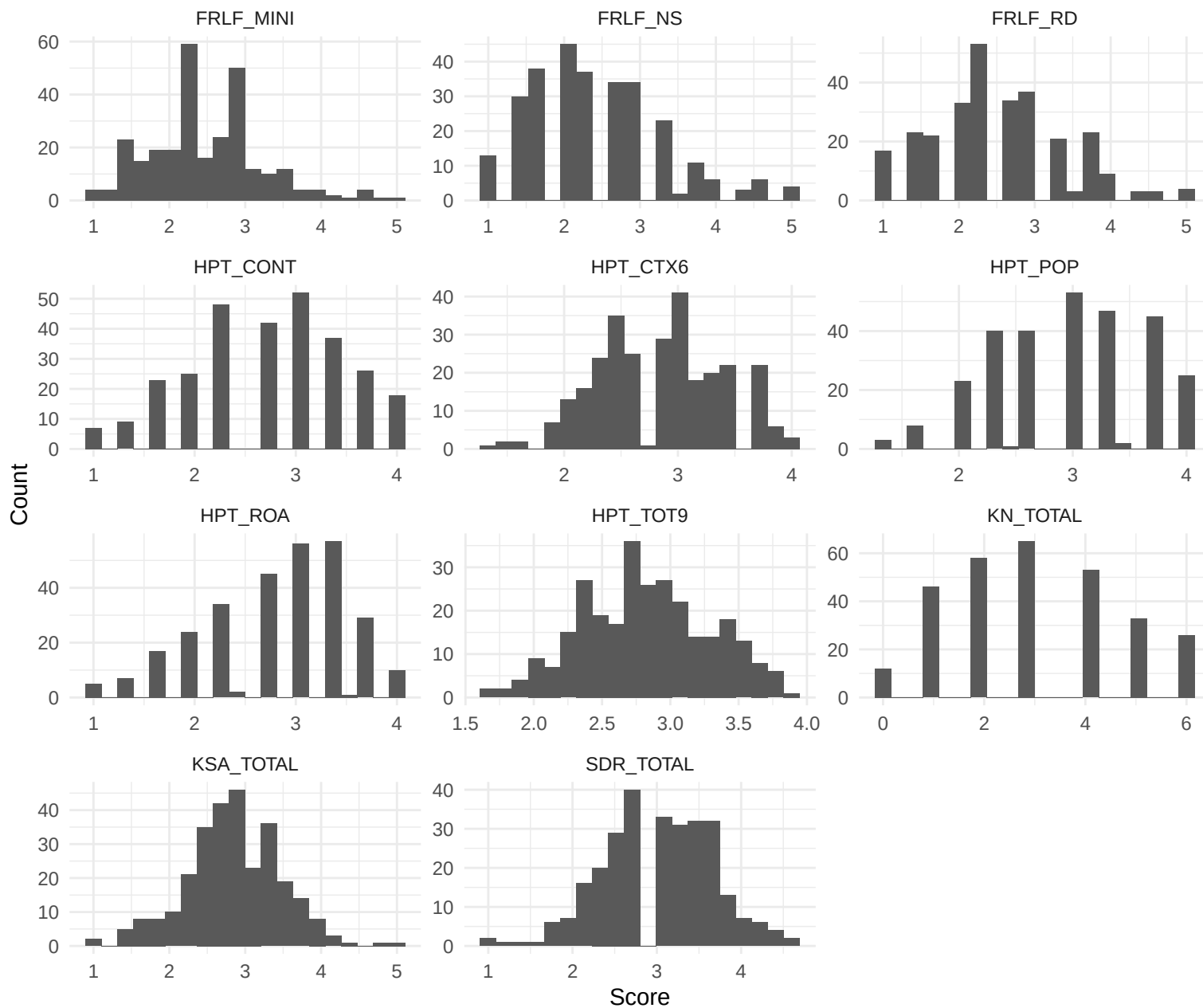
Interpretation: HPT scales should center above 2.5 if students generally avoid presentism and contextualize; FR-LF/KSA are not “good/bad” in isolation but provide ideological context for later modeling; SDR alerts to response bias.

1.5 5) Distributions (histograms, same axes where possible)

```
long_scales <- dat %>%
  select(all_of(desc_vars)) %>%
  pivot_longer(everything(), names_to = "scale", values_to = "value")

ggplot(long_scales, aes(x = value)) +
  geom_histogram(bins = 20) +
  facet_wrap(~scale, scales = "free", ncol = 3) +
  labs(title = "Distributions of scales", x = "Score", y = "Count")
```

Distributions of scales



Interpretation: Watch for spikes at bounds (e.g., KN at 0 or max), floor/ceiling on FR-LF/KSA, and skew in HPT subscores—useful cues for later

transformation or robust modeling.

1.6 6) Zero-order correlations (student level)

```
corr_df <- dat %>% select(all_of(desc_vars)) %>% drop_na()
ct <- psych::corr.test(corr_df, use = "pairwise")
round(ct$r, 2)
```

##	HPT_POP	HPT_ROA	HPT_CONT	HPT_CTX6	HPT_TOT9	KN_TOTAL	FRLF_RD	FRLF_NS
## HPT_POP	1.00	0.16	0.26	0.76	0.64	0.32	-0.09	-0.07
## HPT_ROA	0.16	1.00	0.37	0.34	0.72	0.27	0.02	-0.03
## HPT_CONT	0.26	0.37	1.00	0.82	0.78	0.22	-0.02	0.05
## HPT_CTX6	0.76	0.34	0.82	1.00	0.90	0.34	-0.07	-0.01
## HPT_TOT9	0.64	0.72	0.78	0.90	1.00	0.37	-0.04	-0.02
## KN_TOTAL	0.32	0.27	0.22	0.34	0.37	1.00	-0.06	-0.11
## FRLF_RD	-0.09	0.02	-0.02	-0.07	-0.04	-0.06	1.00	0.41
## FRLF_NS	-0.07	-0.03	0.05	-0.01	-0.02	-0.11	0.41	1.00
## FRLF_MINI	-0.09	0.00	0.01	-0.05	-0.03	-0.10	0.84	0.84
## KSA_TOTAL	-0.13	0.07	0.07	-0.03	0.01	0.05	0.47	0.40
## SDR_TOTAL	-0.05	0.01	0.01	-0.02	-0.01	0.03	-0.07	-0.25
##	FRLF_MINI	KSA_TOTAL	SDR_TOTAL					
## HPT_POP	-0.09	-0.13	-0.05					
## HPT_ROA	0.00	0.07	0.01					
## HPT_CONT	0.01	0.07	0.01					
## HPT_CTX6	-0.05	-0.03	-0.02					
## HPT_TOT9	-0.03	0.01	-0.01					
## KN_TOTAL	-0.10	0.05	0.03					
## FRLF_RD	0.84	0.47	-0.07					
## FRLF_NS	0.84	0.40	-0.25					
## FRLF_MINI	1.00	0.52	-0.19					
## KSA_TOTAL	0.52	1.00	-0.15					
## SDR_TOTAL	-0.19	-0.15	1.00					

```
corr_tab <- as.data.frame(round(ct$r, 2))
corr_tab$Var1 <- rownames(corr_tab)
corr_tab <- corr_tab %>% relocate(Var1)
```

```
kable(corr_tab, booktabs = TRUE, caption = "Zero-order Pearson correlations among constructs (pairwise).") %>%
  kable_styling(latex_options = c("hold_position"))
```

Table 3: Zero-order Pearson correlations among constructs (pairwise).

	Var1	HPT_POP	HPT_ROA	HPT_CONT	HPT_CTX6	HPT_TOT9	KN_TOTAL	FRLF_RD	FRLF_NS	FRLF_MINI
HPT_POP	HPT_POP	1.00	0.16	0.26	0.76	0.64	0.32	-0.09	-0.07	-0.09
HPT_ROA	HPT_ROA	0.16	1.00	0.37	0.34	0.72	0.27	0.02	-0.03	0.00
HPT_CONT	HPT_CONT	0.26	0.37	1.00	0.82	0.78	0.22	-0.02	0.05	0.01
HPT_CTX6	HPT_CTX6	0.76	0.34	0.82	1.00	0.90	0.34	-0.07	-0.01	-0.05
HPT_TOT9	HPT_TOT9	0.64	0.72	0.78	0.90	1.00	0.37	-0.04	-0.02	-0.03
KN_TOTAL	KN_TOTAL	0.32	0.27	0.22	0.34	0.37	1.00	-0.06	-0.11	-0.10
FRLF_RD	FRLF_RD	-0.09	0.02	-0.02	-0.07	-0.04	-0.06	1.00	0.41	0.84
FRLF_NS	FRLF_NS	-0.07	-0.03	0.05	-0.01	-0.02	-0.11	0.41	1.00	0.84
FRLF_MINI	FRLF_MINI	-0.09	0.00	0.01	-0.05	-0.03	-0.10	0.84	0.84	1.00
KSA_TOTAL	KSA_TOTAL	-0.13	0.07	0.07	-0.03	0.01	0.05	0.47	0.40	0.52
SDR_TOTAL	SDR_TOTAL	-0.05	0.01	0.01	-0.02	-0.01	0.03	-0.07	-0.25	-0.19

Interpretation: Correlations locate broad relationships your hypotheses rely on. For example, if **FRLF_MINI** correlates positively with **HPT_CTX6/HPT_TOT9**, this suggests possible ideological alignment inflating contextualization—an effect to test with controls in models (knowledge, SDR) and with item-level checks later.

1.7 7) Between-group variation: school/class ICCs

We estimate the variance attributable to schools and classes with random-intercept models. ICC = proportion of total variance that is between clusters.

```
fit_icc <- function(v) {
  if (!v %in% names(dat)) return(NULL)
  formula <- as.formula(paste0(
    v, " ~ 1 + (1 | school_id) + (1 | school_id:class_label)"
  ))
  tryCatch(
    lme4::lmer(formula, data = dat, REML = TRUE, na.action = na.omit),
    error = function(e) NULL
  )
}

extract_icc <- function(fm) {
  empty <- list(
    ICC_school = NA_real_, ICC_class_within_school = NA_real_,
    ICC_total_cluster = NA_real_, N = NA_integer_, n_schools = NA_integer_,
    n_classes = NA_integer_
  )
  if (is.null(fm)) return(empty)

  variances <- as.data.frame(lme4::VarCorr(fm))
  variance_for <- function(group) {
```

```

    value <- variances$vcov[variances$grp == group]
    if (length(value) == 1L) value else NA_real_
  }
  school_variance <- variance_for("school_id")
  class_variance <- variance_for("school_id:class_label")
  residual_variance <- variance_for("Residual")
  total_variance <- school_variance + class_variance + residual_variance
  flist <- lme4::getME(fm, "flist")

  list(
    ICC_school = school_variance / total_variance,
    ICC_class_within_school = class_variance / total_variance,
    ICC_total_cluster = (school_variance + class_variance) / total_variance,
    N = as.integer(stats::nobs(fm)),
    n_schools = nlevels(flist$school_id),
    n_classes = nlevels(flist[["school_id:class_label"]])
  )
}

targets <- c("HPT_CTX6", "HPT_TOT9", "HPT_POP", "HPT_ROA", "HPT_CONT",
            "FRLF_MINI", "KSA_TOTAL", "KN_TOTAL", "SDR_TOTAL")

icc_rows <- purrr::map(targets, function(sc) {
  components <- extract_icc(fit_icc(sc))
  data.frame(
    scale = sc,
    ICC_school = round(components$ICC_school, 3),
    ICC_class_within_school = round(components$ICC_class_within_school, 3),
    ICC_total_cluster = round(components$ICC_total_cluster, 3),
    N = components$N,
    n_schools = components$n_schools,
    n_classes = components$n_classes
  )
})
icc_res <- dplyr::bind_rows(icc_rows)

kable(icc_res, booktabs = TRUE,
      caption = "Intraclass correlations from nested school/class models.") %>%
  kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)

```

Table 4: Intraclass correlations from nested school/class models.

scale	ICC_school	ICC_class_within_school	ICC_total_cluster	N	n_schools	n_classes
HPT_CTX6	0.041	0.000	0.041	287	10	20
HPT_TOT9	0.085	0.000	0.085	287	10	20
HPT_POP	0.025	0.000	0.025	287	10	20
HPT_ROA	0.092	0.011	0.104	287	10	20
HPT_CONT	0.026	0.007	0.033	287	10	20
FRLF_MINI	0.055	0.015	0.070	284	10	20
KSA_TOTAL	0.057	0.042	0.099	283	10	20
KN_TOTAL	0.053	0.000	0.053	293	10	20
SDR_TOTAL	0.000	0.000	0.000	283	10	20

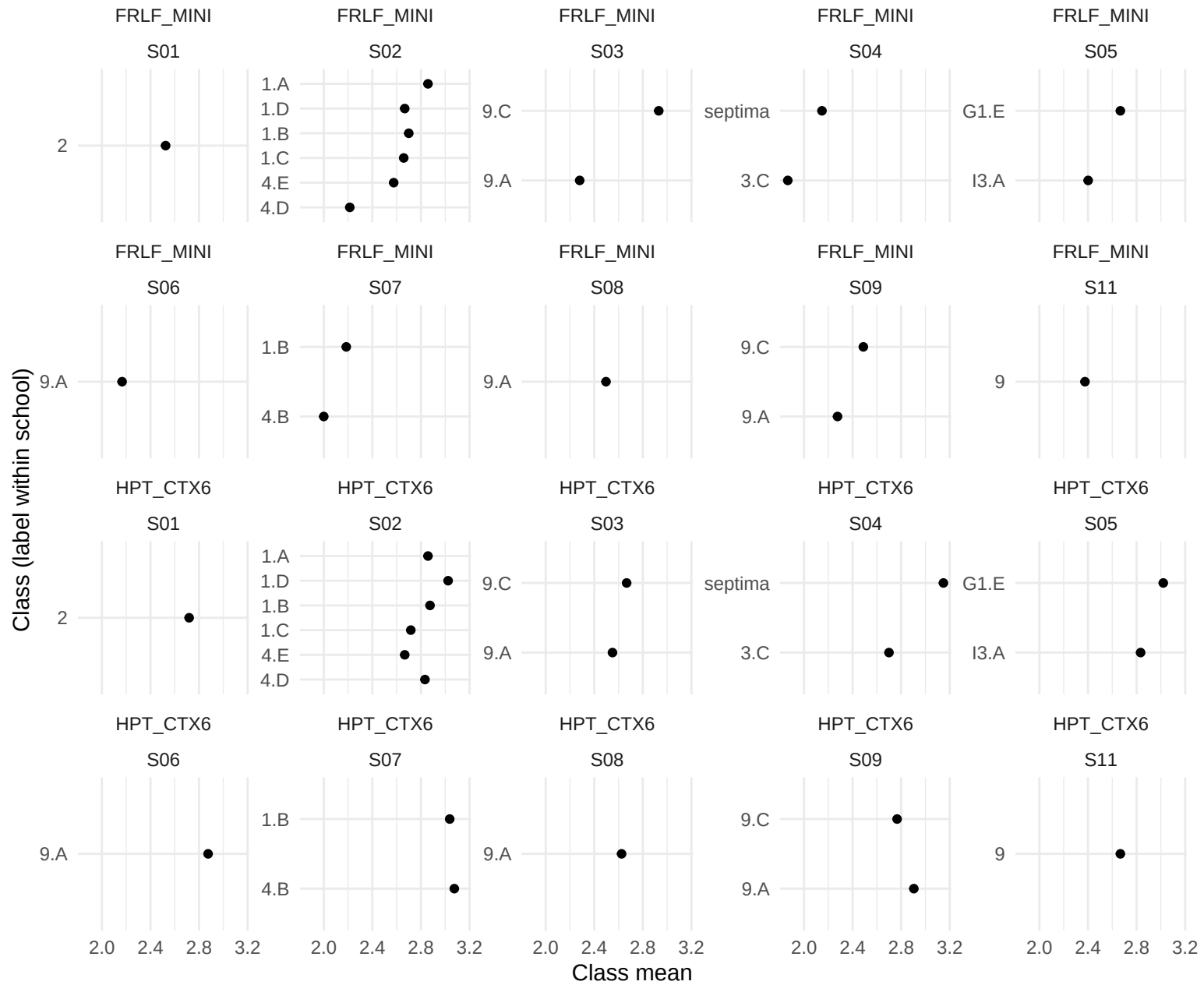
Interpretation: • **ICC ~ 0.00-0.05** → little clustering (ordinary regression OK). • **ICC ~ 0.05-0.15** → modest clustering; use cluster-robust SE or multilevel models. • **ICC > 0.15** → strong clustering; multilevel modeling recommended. Compare **HPT** vs **FR-LF/KSA**: larger ICCs for ideology might reflect school milieu; larger ICCs for HPT may signal classroom task/context effects.

1.8 8) Quick sanity plots by class/school (optional)

```
p1 <- dat %>%
  group_by(school_id, class_id, class_label) %>%
  summarise(n = n(),
            HPT_CTX6 = mean(HPT_CTX6, na.rm=TRUE),
            FRLF_MINI = mean(FRLF_MINI, na.rm=TRUE),
            .groups = "drop") %>%
  pivot_longer(c(HPT_CTX6, FRLF_MINI), names_to="scale", values_to="mean") %>%
  ggplot(aes(x = reorder(class_label, mean), y = mean)) +
  geom_point() +
  facet_wrap(scale ~ school_id, scales = "free_y") +
  coord_flip() +
  labs(x = "Class (label within school)", y = "Class mean",
       title = "Class means within schools (HPT vs FR-LF)")
```

p1

Class means within schools (HPT vs FR-LF)



Interpretation: Visual check for unusually high/low classes can inform later sensitivity checks (e.g., re-running models without extreme classes).

1.9 9) Reproducibility appendix

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0 LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=cs_CZ.UTF-8      LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=cs_CZ.UTF-8  LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=cs_CZ.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=cs_CZ.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Prague
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
##  [1] lme4_2.0-6      Matrix_1.7-6    psych_2.6.5     scales_1.4.0
##  [5] ggplot2_4.0.3   kableExtra_1.4.1 knitr_1.51      readxl_1.5.0.1
##  [9] purrr_1.2.2     tidyr_1.3.2     stringr_1.6.0   dplyr_1.2.1
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4    xml2_1.6.0      stringi_1.8.7    lattice_0.23-1
##  [5] digest_0.6.39     magrittr_2.0.5  evaluate_1.0.5    grid_4.6.1
##  [9] RColorBrewer_1.1-3 fastmap_1.2.0    cellranger_1.1.0  tinytex_0.61
## [13] viridisLite_0.4.3 textshaping_1.0.5 reformulas_0.4.4  Rdpack_2.6.6
## [17] mnormt_2.1.2      cli_3.6.6       rlang_1.3.0      rbibutils_2.4.1
## [21] splines_4.6.1     withr_3.0.3     yaml_2.3.12      otel_0.2.0
## [25] tools_4.6.1       parallel_4.6.1  nloptr_2.2.1     minqa_1.2.8
## [29] boot_1.3-32       vctrs_0.7.3     R6_2.6.1         lifecycle_1.0.5
## [33] MASS_7.3-66       pkgconfig_2.0.3 pillar_1.11.1     gtable_0.3.6
## [37] Rcpp_1.1.2        glue_1.8.1      systemfonts_1.3.2 xfun_0.60
## [41] tibble_3.3.1      tidyselect_1.2.1 rstudioapi_0.19.0 farver_2.1.2
```

## [45]	htmltools_0.5.9	nlme_3.1-171	labeling_0.4.3	rmarkdown_2.32
## [49]	svglite_2.2.2	compiler_4.6.1	S7_0.2.2	

1.9.1 Notes & interpretation pointers

- **HPT scales (1-4):** Higher indicates better contextualization/agent-sensitive reasoning—**after POP reversal**; we report CTX6 (primary) and TOT9 (with ROA) side-by-side.
- **FR-LF mini (1-5):** Short right-wing authoritarian/Nazi relativization composite; higher = stronger endorsement. Use primarily as a predictor/covariate and for DIF checks later.
- **Knowledge:** Treat as a covariate; it often shows small but non-zero links to HPT.
- **SDR:** Use to check attenuation/amplification of sensitive attitudes.